

Linux Programming: Assignment-3: 20-09-2025

1. Distinguish between **man** and **whatis commands**? Justify with proper example.
(CO1)
2. Use the **tee command** to save the output of **ls -l** into a file while also displaying it. (CO4)
3. Explain with an example how the **tee command** can be used in **logging**. (CO4)
4. List the steps involved in installing **Ubuntu 25.04 LTS** on Oracle VirtualBox.
(CO2)
5. During **Ubuntu OS** installation, you face a **Kernel Panic Error**. How would you troubleshoot it? (CO3)
6. Write the **command** to display the system's **hostname**? How to change hostname using **sysctl** command? (CO1)
7. Which command is used **to show the calendar of the year 1984** with **August** month? (CO1)
8. Write a command **to display system uptime and logged-in users together**.
(CO3)
9. Use the **find command** to list all **".c" files in /home/user**. (CO1)
10. How **do you change file permissions** to allow only the **owner** to read and write?
(CO1)

Linux Programming: Assignment-4: 20-09-2025

1. A system has **a file /etc/passwd**. How would you use **grep + tee** to extract **usernames** and save them to a file while also displaying them on screen? (CO4)
2. A **binary isn't found in \$PATH**. How would you use **commands** (which, find, locate) to troubleshoot and fix the issue? (CO3)
3. Write a **command pipeline** that finds all **.log files** modified in the last 24 hours in **/var/log** and saves results into **log_report.txt**. (CO4)
4. What is the difference between **shutdown -r now** and **reboot**? (CO1)
5. How can you use the **tee command** to debug a script that generates both standard output and error messages? (CO4)
6. Explain any **three real-world applications of Linux** in industries. (CO2)
7. Differentiate application, system and utility software in the context of Linux environment. (CO2)
8. What are the key differences **between open-source and proprietary operating systems**? (CO2)
9. Write the command to display the **system's kernel version**. (CO3)
10. What is the difference between **head** and **tail** commands in text processing? (CO1)

Linux Programming: Assignment-5: 20-09-2025

1. What is a **shell** in Linux OS? How many categories of **shell** is currently exists in Linux? Why **bash shell** is very popular in Linux distribution? (CO2)
2. What does the **ls -Z** command display? (CO1)
3. Write a command to list **all hidden files** in the current directory. (CO1)
4. Explain the **difference between hard links and soft links** (symbolic links) in Linux. (CO1)
5. A file has permissions **-rwxr-x--x**. Explain who can read, write, and execute it. (CO1)
6. Write the command to **change the group ownership** of a file data.txt to group staff. (CO1)
7. Why is it dangerous to give **777 permissions to a file**? Explain with an example. (CO1)
8. What is the difference between **apropos** (i.e., **man -k**) and **whatis** (i.e., **man -f**)? (CO1)
9. Write a command to **redirect the error output of a command** to a file named **error.log**. (CO1)
10. How can you use the **tee command** to append output to a file instead of overwriting it? (CO4)

Linux Programming: Assignment-6: 20-09-2025

1. Which **command** is used **to list the contents of a directory**? Justify with proper example. (CO1)
2. Write the **command to create a new directory named 123test_dir**. (CO1)
3. What is the purpose of the **sed** command? Justify with proper example. (CO1)
4. Which **distinct command** is used **to display one-line descriptions of any commands**? (CO1)
5. Write the command to **create** an empty file named **“notes.txt”**. (CO1)
6. Differentiate between **grep** and **awk** commands with an example. (CO2)
7. Write the command to give **read, write, and execute permission** to the owner of a file script.sh. (CO1)
8. How is **chown** different from **chgrp**? Give one example for each. (CO1)
9. A user complains that they **cannot execute a file** even though it exists in their directory. How would you troubleshoot this using **ls -l, chmod, and whoami**? (CO3)
10. Design a command pipeline to: **find all .log files modified in the last 2 days** in /var/log, display them on screen, and **save the results into a file recent_logs.txt** using **tee command**. (CO4)

Note:

- (i) All the above **40 questions** (**Assignment-3 to Assignment-6**) are from **Module-1 and Module-2**. Refer your Textbook or Reference book to answer all. You can use **Central Library Books, and Open AI Tools** also. But your answer must be decorated in your own language.
- (ii) We will accept **softcopy** (.docx file) from your end but it will go through content similarity checking through **TURNITIN** Software. So, don't try to copy paste from the **Internet**. If we find similarity in the content, your **Internal Marks** will be less.
- (iii) Each question carries 03 Marks: **Total 120 Marks. Submission Deadline: 05th October, 2025.**
- (iv) Try to submit all your **assignment** as per the **deadline**. If you miss the deadline your **Internal marks** will be reduced.

CIA Marks Distribution:

- 1. MSE (01): 30 marks.
- 2. Assignments (10 nos.): 30 marks [**Be careful with assignments**].
- 3. Quizzes (04 nos.): 20 marks.
- 4. Individual Activity (04 nos.): 10 marks.
- 5. Group Activity (01): 10 marks.